

Blister Bot: Master Software Architecture & Logic Flow

System Description for AI Implementation

This document outlines the complete state machine, hardware interactions, and database workflows for the Blister Bot edge-AI system.

State 0: Boot Sequence & Hardware Initialization

Upon system power-on (Arduino Uno Q / Edge Device), the system executes a pre-flight check.

1. **Peripheral Check:** Ping and verify connections to Camera 1 (Face/User), Camera 2 (Meds/Wheel), and the Type-C Microphone.
2. **Audio Check:** Initialize Bluetooth module and pair with the external speaker.
3. **Boot Confirmation:** Trigger Text-to-Speech (TTS) via speaker: *"Hi, I'm awake."*
4. **Transition:** Listen for the wake word trigger: *"bot , setup"* .

State 1: Setup Mode (Part 1 - User Profiles)

Triggered by the user saying *"bot , setup"* .

1. **Input Selection:** System asks (TTS): *"Phone or voice?"* and waits for Mic input.
 - *Branch A (Phone):* [Placeholder for Future App Integration]
 - *Branch B (Voice):* Proceed to conversational setup.
2. **The Caretaker Loop:**
 - System asks for Caretaker details.
 - **Data required:** Name, Work Shift (Morning/Evening/Full Day), and an `is_patient` boolean (is the caretaker also a patient?).
 - **Parallel Process (Cam 1):** While the user speaks, Cam 1 captures facial data and generates an embedding for Face ID.
 - **Condition:** If `is_patient == True` , branch immediately to the Patient Loop, then return here.
 - **Loop Check:** System asks: *"Are there any more caretakers to add?"* (Yes = Repeat loop, No = Proceed to Patient Loop).
3. **The Patient Loop:**
 - System asks (TTS): *"Enter patient name"* and proceeds to ask for health details.
 - **Data required:** Name, Age, Gender, Health Issues, Medicine Schedule (Morning/Evening, Pre-food/Post-food), and Assigned Caretaker.
 - **Parallel Process (Cam 1):** Scan patient face and generate Face ID embedding.
 - **LLM Entity Extraction:** The user can answer in natural, conversational sentences. The LLM parses the transcript, extracts exact variables, and structures them into JSON.
 - **Database Write:** Commit structured JSON to `Users` , `Caretakers` , and `Patients` SQLite tables.
 - **Loop Check:** System asks: *"Are there more patients?"* (Yes = Repeat, No = End Part 1).

State 2: Setup Mode (Part 2 - Medication Inventory)

1. **Physical Loading:** User physically inserts blister packs into the wheel.

2. **Confirmation:** User verbally confirms all meds are loaded.
3. **Computer Vision (Cam 2) OCR Scan:**
 - Wheel rotates to each slot.
 - Cam 2 scans the blister pack text.
 - Extracts: Medicine Name, Dosage.
4. **Validation Pipeline:**
 - Cross-reference OCR results with an internal/local medical database to verify drug details and correct partial/blurry OCR reads.
5. **Database Write:**
 - Store medication metadata and inventory counts (slots filled) in the `Inventory` SQLite table.
6. **Transition:** Setup mode complete. Transition to Normal Operating Mode.

State 3: Normal Operating Mode (The Main Loop)

A. The Alert & Authentication Phase

1. **Time-Based Trigger:** System checks the SQLite schedule. When a med time arrives, trigger a TTS reminder/alarm via the speaker.
2. **Presence Detection:** When a person approaches the camera or speaks:
 - Cam 1 / Mic authenticates the user via Face ID or Voice Print.
3. **Contextual Interaction (LLM):**
 - *If User == Patient:* Check schedule rules. Ask, "*Have you eaten?*" for post-food meds, or vice versa for pre-food meds.
 - If conditions fail (e.g., hasn't eaten), TTS instructs them to eat and return. Halt dispensing.
 - *If User == Caretaker:* Ask which patient they are collecting for, then proceed with the same condition checks for that specific patient.

B. The CNC Dispensing Phase

Once interaction conditions are satisfied:

1. **Indexing:** Rotate the medicine wheel (MG995/Stepper) until the specific patient's medication is facing the gantry.
2. **Vision Targeting (Cam 2 + Meta SAM 2):**
 - Cam 2 takes an image of the blister pack.
 - SAM 2 (Segment Anything Model) identifies the exact boundary/outline of the target pill cavity.
 - Calculate G-code/motor steps for this specific boundary.
3. **Cutting Sequence (CNC Gantry):**
 - **Plunge:** Z-axis motor thrusts the needle down into the boundary.
 - **Cut:** X and Y axis motors trace the SAM 2 boundary, cutting the foil/plastic.
4. **Ejection:**
 - Wheel rotates downward precisely enough so gravity pulls the cut pill/blister chunk out of the machine.



- Repeat sequence if multiple pills are scheduled.

C. Inventory Update & Alerts

1. **Database Update:** Decrement the count for the dispensed medication in the SQLite `Inventory` table.
2. **Low Stock Alert:**
 - If `med_count == 1` : Immediately trigger a TTS warning via speaker.
 - Log a reminder to play this low-stock alert again right before the *next* scheduled dispensing timeslot.